INTERNET OF THINGS

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WHAT IS INTERNET OF THINGS?

- The Internet of Things (IoT) refers to a network of physical objects "things“ embedded with sensors, software, and technologies that enable them to connect, collect, and exchange data with other devices and systems over the internet or networks. These smart, interconnected devices range from household appliances to industrial tools, automating tasks and enabling remote monitoring and, in many cases, autonomous operation.

IoT enhances operational efficiency and decision-making by allowing for real-time data analysis and automation without manual intervention.

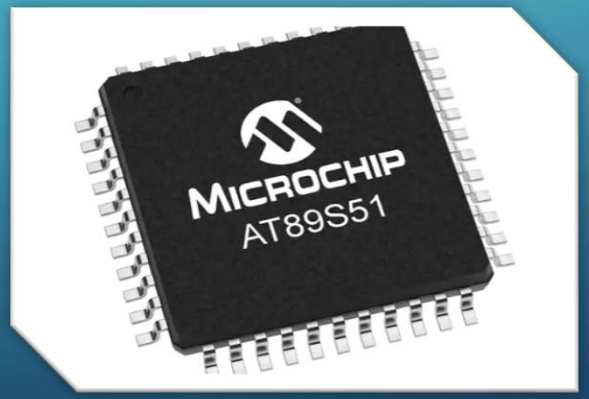
KEY ASPECTS OF IOT

- **Components:** IoT systems combine physical devices (sensors/actuators), internet/network connectivity, data processing, and user interfaces.
- **Functionality:** They collect data from their environment, transmit it, and often act on it to improve efficiency, convenience, or automation.
- **Examples:** Smart thermostats, connected vehicles, wearable fitness trackers, smart home security systems, and industrial sensors.
- **Applications:** Commonly used for smart homes, healthcare, industrial IoT and smart cities.

MICROCONTROLLERS

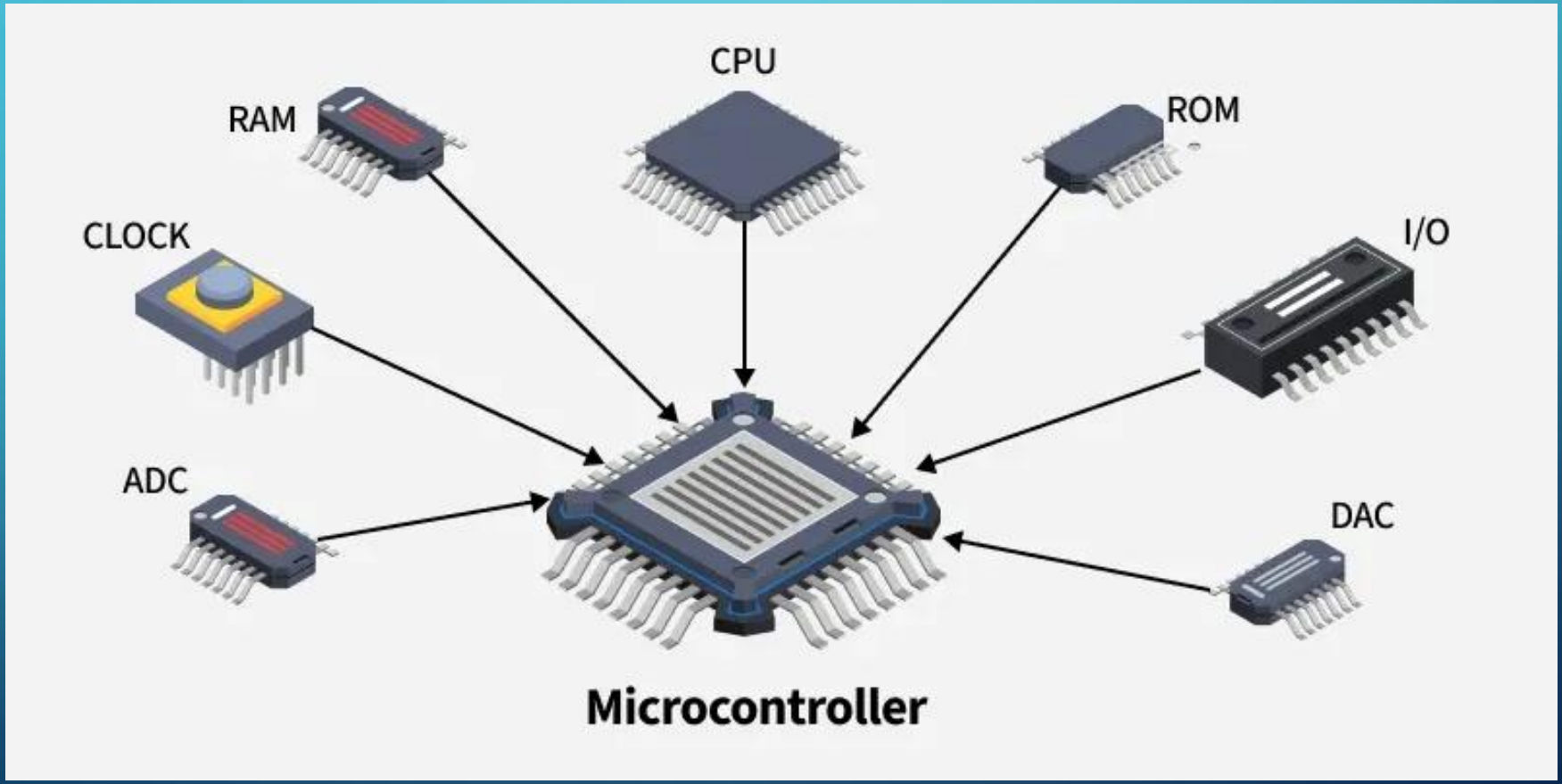
- A microcontroller (MCU) is a small, self-contained computer on a single chip, integrating a processor, memory (RAM, ROM/Flash), and programmable input/output (I/O) peripherals, designed to execute specific tasks in embedded systems like home appliances, cars, and IoT devices







WHAT'S INSIDE A MICROCONTROLLER?



WHAT'S INSIDE A MICROCONTROLLER?

- CPU – runs your program
- Memory – stores code and variables
- Input/Output pins (GPIO) – talk to the outside world
- Timers / counters
- Peripherals – ADC (analog reading), UART, I^2C, SPI, PWM, etc.

WHAT IT IS NOT (MICROCONTROLLERS)

- ❌ Not a PC
- ❌ No keyboard, mouse, monitor, or operating system (usually)
- ❌ Not meant for multitasking heavy apps

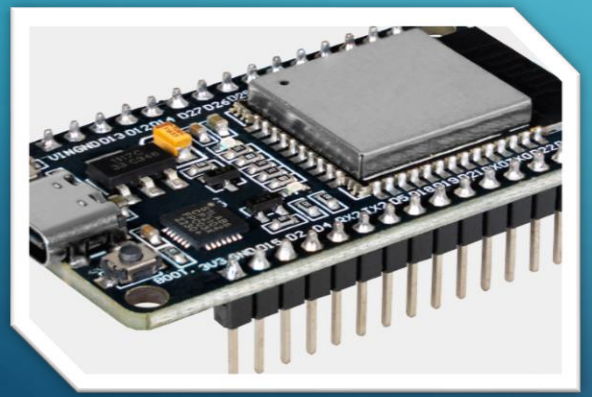
It runs **one dedicated job**, usually forever, in a loop.

WHERE CAN YOU FIND MICROCONTROLLERS?

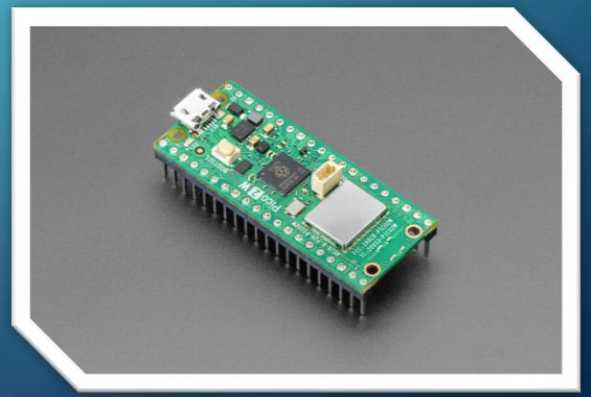
- Washing machines
- Microwave ovens
- TV remotes
- Elevators
- Cars
- Robots
- IoT devices
- Keypads

MICROCONTROLLER DEVELOPMENT BOARDS

- A microcontroller development board (dev board) is a ready-to-use board built around a microcontroller that makes it easy to power, program, and test your code without designing hardware from scratch.







ESP32 DEV BOARD

- An ESP32 development board is a versatile, low-cost circuit board featuring Espressif's powerful ESP32 Wi-Fi and Bluetooth System-on-Chip (SoC), designed for IoT, embedded projects, and prototyping

ESP32 VS ARDUINO?

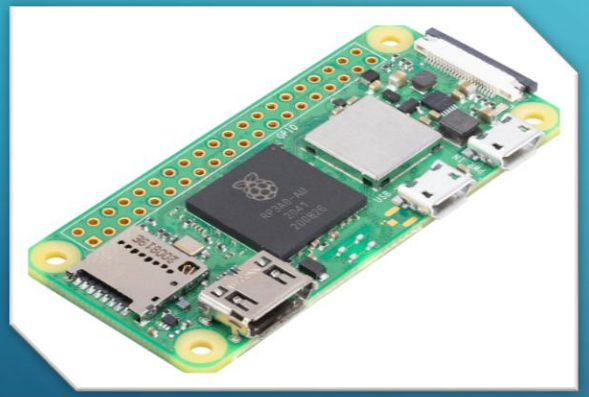
Feature	Arduino	ESP32
Small IoT projects	✅	✅
Wi-Fi / Bluetooth IoT	❌	✅
Battery-operated projects	✅	✅ (with deep sleep)
Advanced projects (AI, voice, sensors, network)	❌	✅
Real-time control (motors, robots)	✅	✅

ESP32 DEV BOARD FAMILY



SINGLE BOARD COMPUTERS (SBC)

- A Single Board Computer (SBC) is a complete computer built on one circuit board that can run a full operating system.







WHAT AN SBC INCLUDES

- SoC (CPU + often GPU)
- RAM
- Storage (microSD / eMMC)
- GPU / video output
- Networking (Ethernet, Wi-Fi, Bluetooth)
- USB ports
- HDMI / Display output
- Audio
- GPIO header

ELECTRONIC COMPONENTS

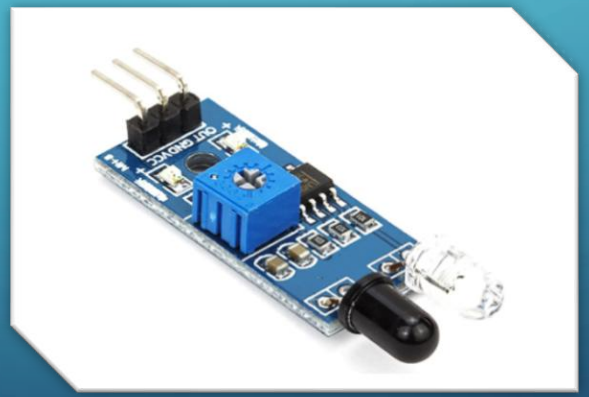
- SENSORS (Input Devices)
- ACTUATORS (Output Devices)
- PASSIVE COMPONENTS
- ACTIVE COMPONENTS

SENSORS

- A sensor detects or measures a physical property and converts it into an electrical signal that a controller can read.
- Sensors = INPUT
- Common things sensors measure: Temperature, Light, Distance, Water level, Sound, Motion / acceleration, Magnetic field, Touch / pressure

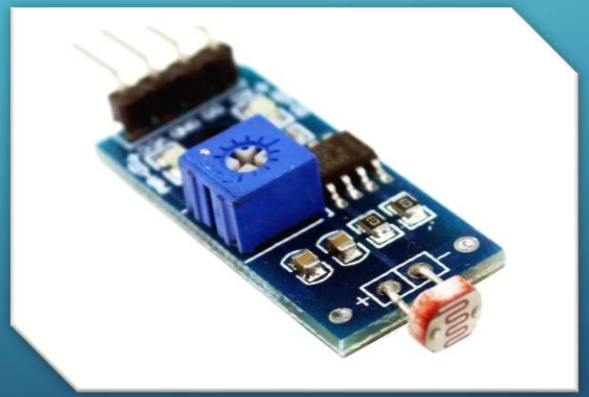
IR PROXIMITY SENSOR

- IR proximity sensor modules are compact and versatile sensors used to detect nearby objects without physical contact by emitting and receiving infrared light.



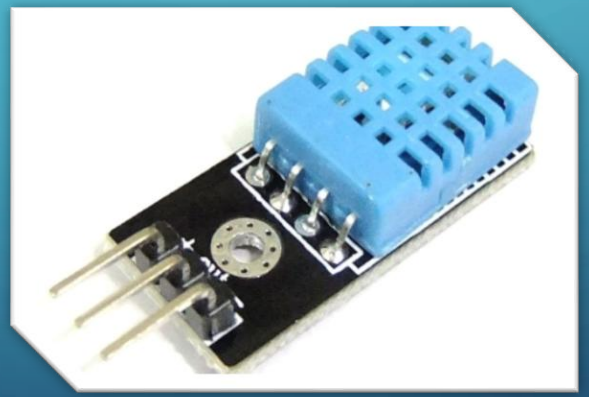
PHOTOSENSITIVE RESISTOR SENSOR

- A photosensitive resistor module is an electronic component featuring a Light Dependent Resistor (LDR) to detect ambient light intensity and brightness



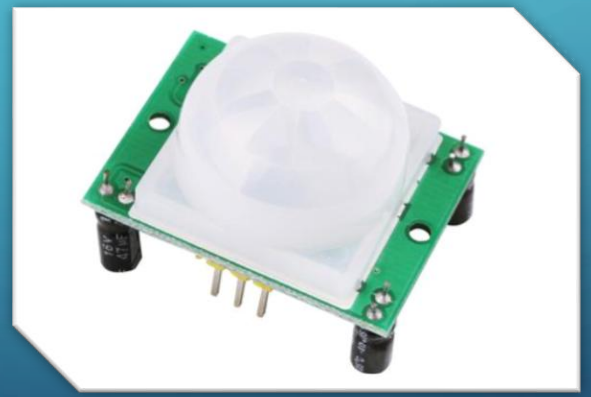
DHT11 TEMPERATURE AND HUMIDITY SENSOR

- The DHT11 sensor module is a digital sensor used for measuring temperature and humidity



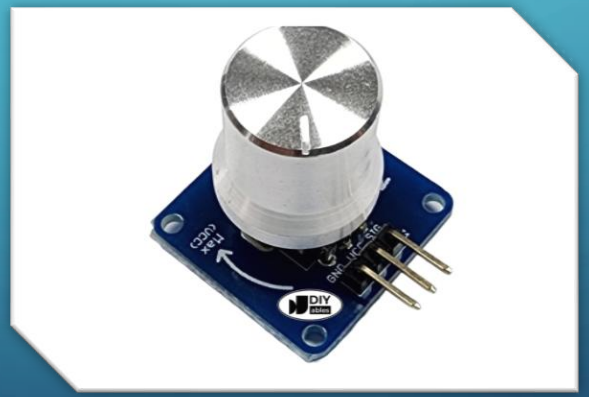
HC-SR501 PIR MOTION SENSOR

- The HC-SR501 is a popular, passive infrared (PIR) motion sensor module designed for detecting human or animal movement within a 3-7 meter range and a 120° angle.



POTENTIOMETER SENSOR

- A potentiometer sensor module is a 3-pin analog device that acts as a variable voltage divider, providing a continuously variable output voltage based on the rotation of a knob.



PUSH BUTTON SENSOR

- A button switch module is an electronic component that detects the state of a button (pressed or released) and is used to connect or break circuits in a wide variety of applications, from doorbells to industrial machinery.



NEO-M8N GPS MODULE SENSOR

- The u-blox NEO-M8N is a popular, high-performance GNSS (Global Navigation Satellite System) module that offers concurrent reception of up to three GNSS systems for enhanced positioning accuracy and reliability.

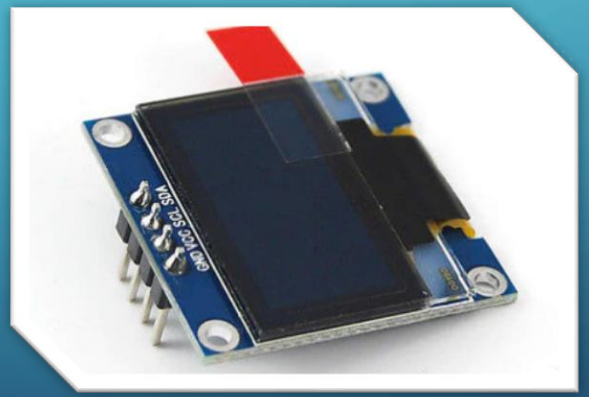


ACTUATORS

- An actuator takes an electrical signal and converts it into physical action.
- Actuators = OUTPUT
- Common actions: Move, Light up, Make sound, Lock / unlock, Heat, Open / close, Display

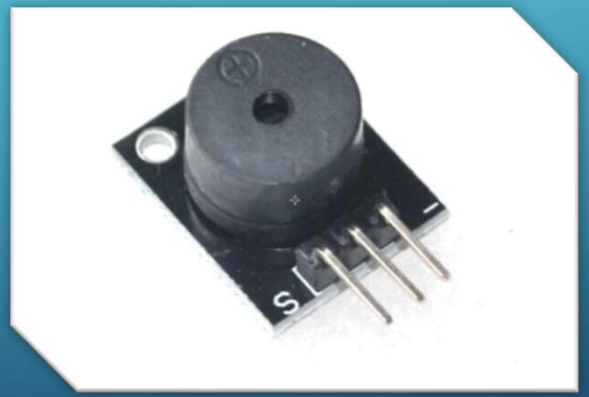
0.96” OLED SCREEN ACTUATOR

- The 0.96” OLED display module is a compact, 128x64 pixel, self-luminous screen commonly used for small-scale, low-power projects (e.g., wearables) with Arduino, Raspberry Pi, or ESP8266.



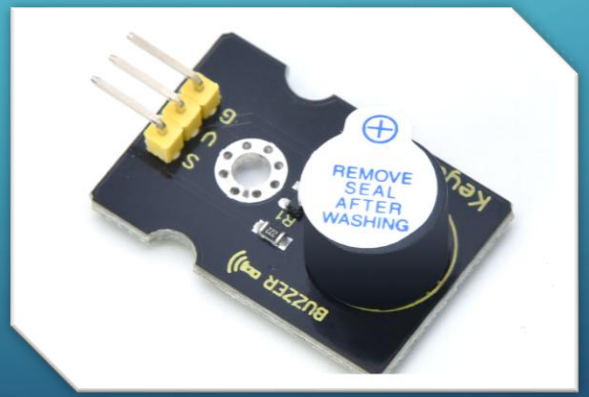
PASSIVE BUZZER ACTUATOR

- A passive buzzer module is a sound-producing component needing an external signal (like from an Arduino) to generate tones, allowing control over pitch and melody by varying the signal's frequency.



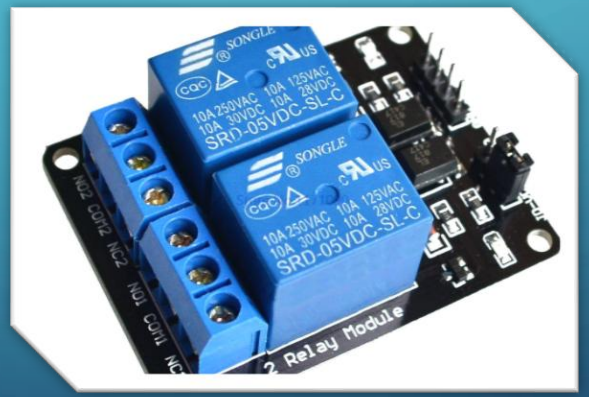
ACTIVE BUZZER ACTUATOR

- An active buzzer module is a simple electronic component with a built-in oscillator that produces a fixed-frequency, single-tone sound as soon as it receives power.



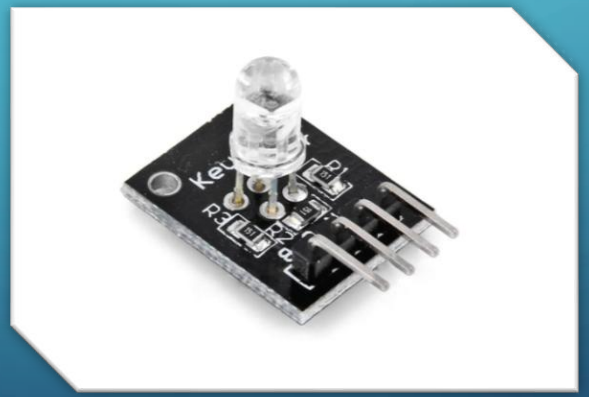
5V 2-CHANNEL RELAY MODULE ACTUATOR

- A 5V 2-channel relay module is an electronic board that uses a low-power 5V signal to control higher voltage/current AC or DC circuits.



LED RGB ACTUATOR

- An RGB LED module is a versatile lighting component, often used with microcontrollers like Arduino, that mixes red, green, and blue light to create a full spectrum of colors.



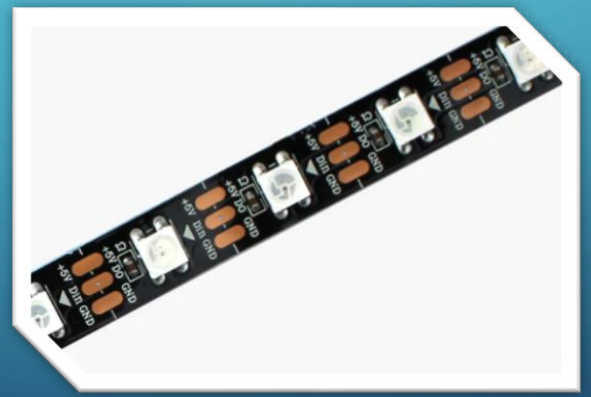
SINGLE COLOR 5MM LED ACTUATOR

- LED modules are widely available as individual components or pre-wired units, often with a built-in resistor for simplified use across a range of voltages. These products are commonly used in DIY electronics projects, indicator lights, and model building.



WS2812 / WS2812B ARGB LEDS

- The WS2812B LEDS is an individually addressable 5V ARGB (Addressable RGB) LED, featuring an integrated IC, allowing each LED to display unique colors and intensities independently.



PASSIVE COMPONENTS

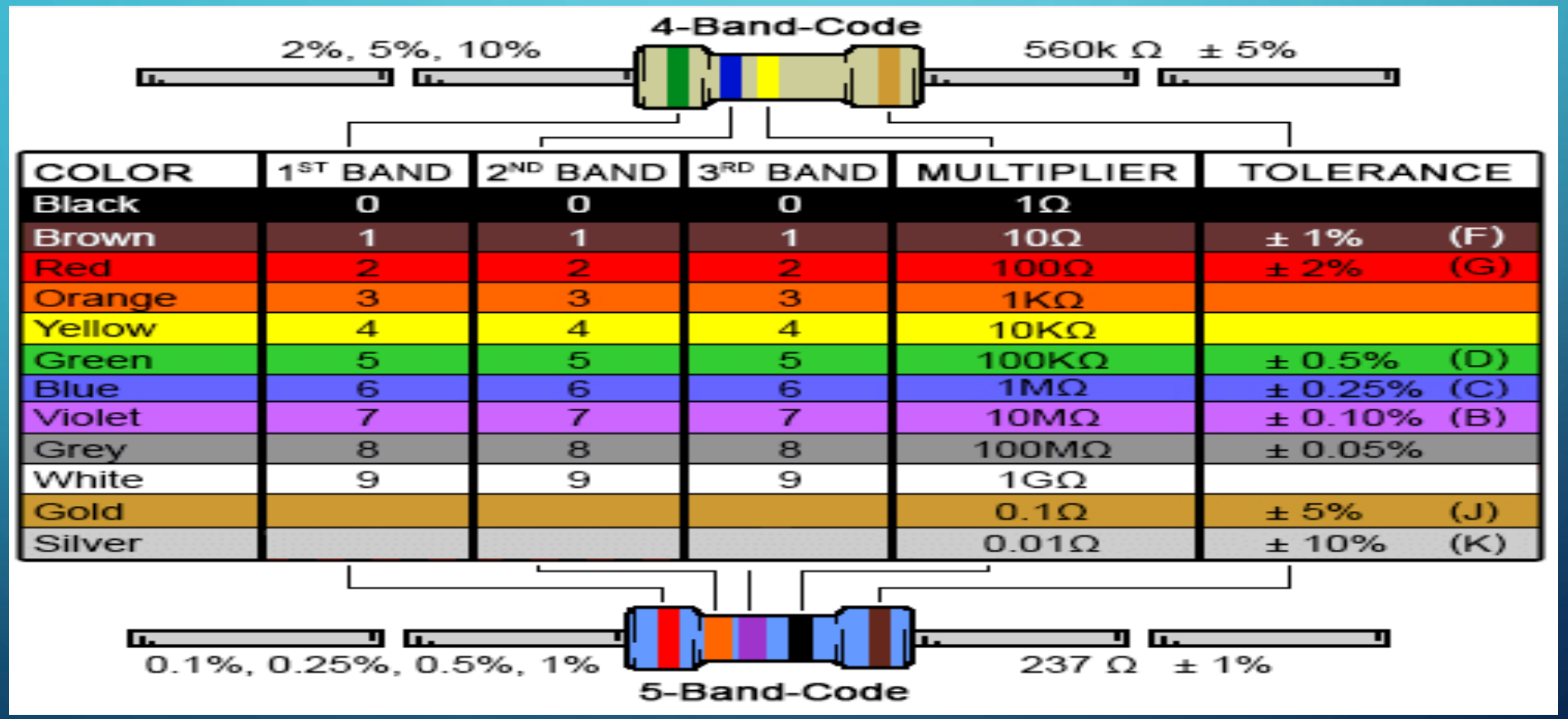
- Components that cannot generate energy themselves; they only store, limit, or shape electrical signals.
- No amplification or control; they just support the circuit.
- Examples: Resistors, Capacitors, and Inductors.

RESISTOR

- A resistor is a passive two-terminal electronic component that implements electrical resistance as a circuit element.



RESISTANCE AND TOLERANCE OF A RESISTOR



COLOR	1ST BAND	2ND BAND	3RD BAND	MULTIPLIER	TOLERANCE
Black	0	0	0	1Ω	
Brown	1	1	1	10Ω	± 1% (F)
Red	2	2	2	100Ω	± 2% (G)
Orange	3	3	3	1KΩ	
Yellow	4	4	4	10KΩ	
Green	5	5	5	100KΩ	± 0.5% (D)
Blue	6	6	6	1MΩ	± 0.25% (C)
Violet	7	7	7	10MΩ	± 0.10% (B)
Grey	8	8	8	100MΩ	± 0.05%
White	9	9	9	1GΩ	
Gold				0.1Ω	± 5% (J)
Silver				0.01Ω	± 10% (K)

CAPACITOR

- A capacitor is a passive, two-terminal electronic component that stores electrical energy in an electric field by accumulating charges on two closely spaced conductive plates separated by a dielectric insulator.



INDUCTORS

- Inductors are passive electrical components, essentially coils of wire, that store energy in a magnetic field when current flows through them, resisting changes in current.

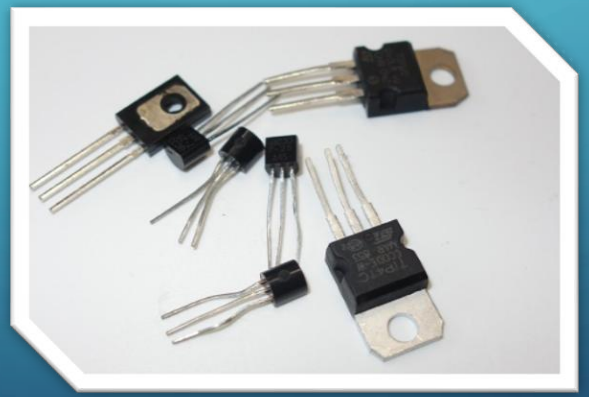


ACTIVE COMPONENTS

- Components that **can control, amplify, or switch electrical signals**, often requiring a power source.
- They actively manipulate signals, unlike passive components.
- Examples: Transistors, Diodes, and Integrated Circuits.

TRANSISTOR

- Transistors are essential semiconductor devices that function as electronic switches or amplifiers, regulating the flow of electricity by allowing a small voltage or current to control a much larger current.



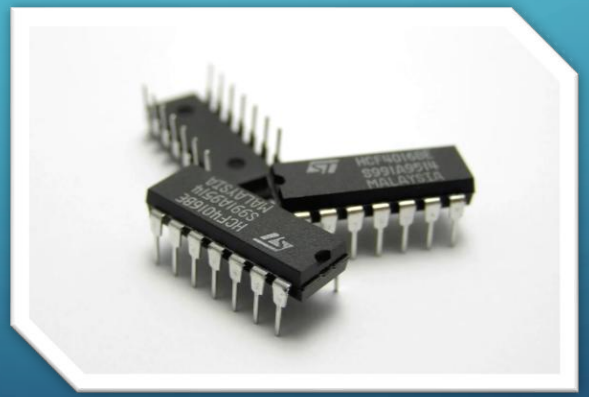
DIODE

- Diodes are two-terminal semiconductor devices that act as one-way valves for electric current.

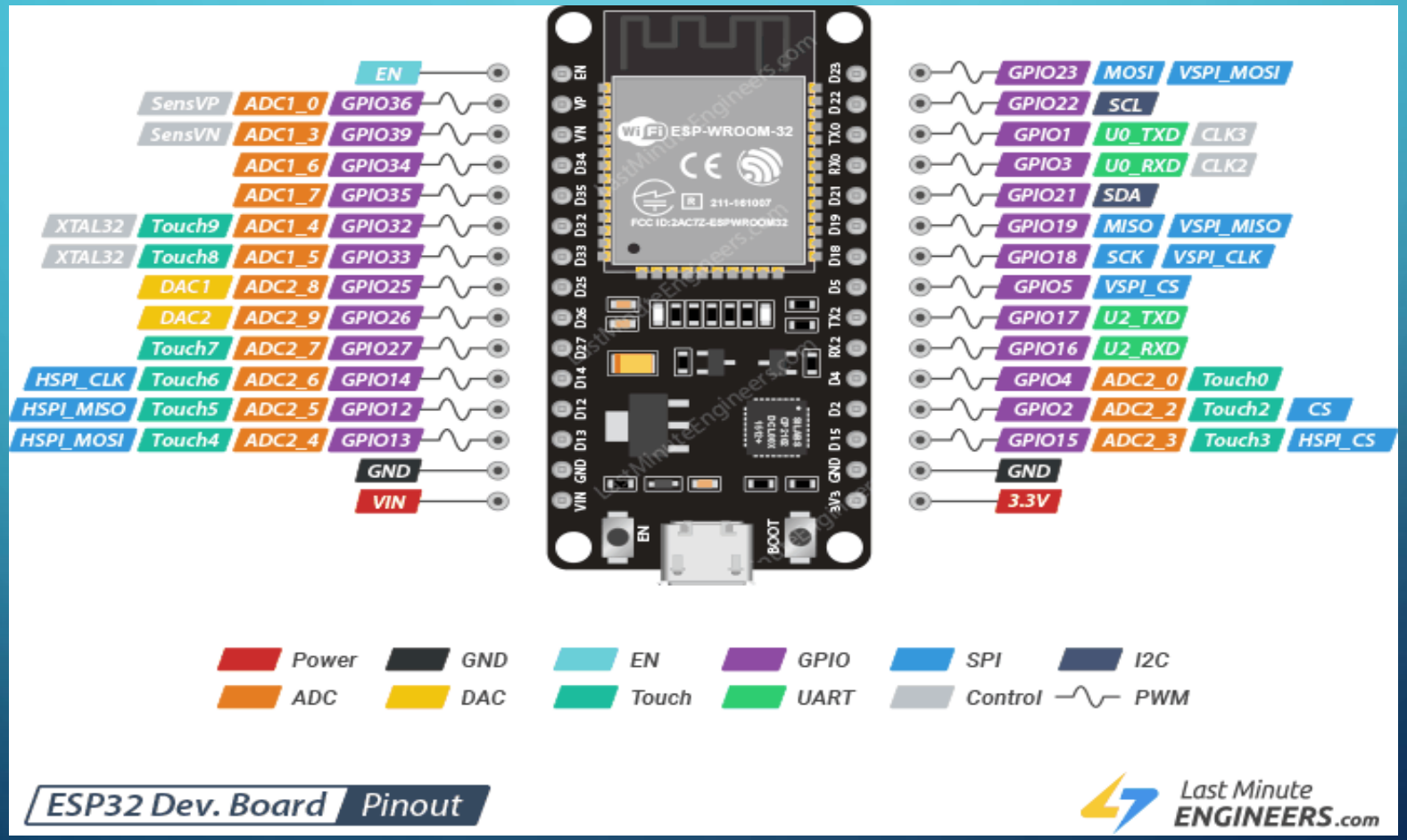


INTEGRATED CIRCUITS

- An integrated circuit (IC), or microchip, is a tiny electronic device that combines many electronic components (like transistors, resistors, and capacitors) onto a single semiconductor chip allowing them to work together to perform complex functions.



ESP32 PINOUT (30 PINS)



WHAT IS GPIO?

- GPIO (General Purpose Input/Output) refers to digital pins on microcontrollers or circuit boards that can be programmed by software to act as either inputs (to receive signals/data) or outputs (to send signals/control devices), allowing them to interface with external components like sensors, LEDs, buttons, or motors, serving as a versatile bridge for communication in embedded systems. These pins have no fixed function until defined by the designer, enabling flexible control for tasks like reading button presses or blinking LEDs.

WHAT IS ADC?

- ADC = Analog-to-Digital Converter
- A component that converts an analog signal (continuous voltage) into a digital number that a microcontroller can understand.
- Microcontrollers cannot directly read analog voltages as numbers. They only understand digital signals (binary 0s and 1s). ADC solves this by measuring the voltage and outputting a digital value.

WHAT IS DAC?

- DAC = Digital-to-Analog Converter
- A component that converts a digital value (from a microcontroller) into an analog voltage or current.
- Microcontrollers output digital signals (HIGH/LOW, 1s and 0s). Some devices require analog signals: Speakers (audio), Motors (analog speed control), Voltage-controlled devices. DAC allows the MCU to control analog devices directly.

WHAT IS UART?

- UART = Universal Asynchronous Receiver/Transmitter
- A hardware module (or protocol) that allows serial communication between devices by sending data one bit at a time asynchronously.
- Serial communication: Data is sent bit by bit over a single wire (plus ground)
- Asynchronous: Sender and receiver do not share a clock — they must agree on a baud rate (speed).
- Common for MCU ↔ MCU or MCU ↔ PC communication
- Rx to Tx then Tx to Rx

WHAT IS SPI?

- SPI = Serial Peripheral Interface
- A high-speed serial communication protocol that lets the microcontroller talk to devices like: SD cards, Displays (OLED, TFT) Sensors, Flash memory
- Faster communication than UART

OTHER PIN INTERFACES

- EN – A pin that turns the microcontroller/module ON or OFF.
- VIN – A pin used to supply power to the development board from an external source (battery, adapter, etc.), usually before voltage regulation.
- GND – Ground: A reference point in the circuit for all voltages, and the return path for electric current.
- 3.3V – A pin that provides a regulated 3.3-volt output from the board's voltage regulator. It can also sometimes be used as a power input on some boards.
- Touch – Special GPIO pins that can detect touch.

PROGRAMMING ESP32 WITH ARDUINO IDE

- Getting Started with the ESP32 Development Board | Random Nerd Tutorials
- Demonstration